# MILITARY SPECIFICATION

## COMPARATOR, VISCOSITY, OIL, 200 - 2000 SAYBOLT SECONDS

### 1. SCOPE

1.1 This specification covers one type of pocket comparator, suitable for rapid determination of the viscosity of light oils in Saybolt seconds at 100° Fahrenheit (F.).

### 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

| | |
|---|---|
| PPP-B-585 | Boxes, Wood, Wirebound. |
| PPP-B-601 | Boxes, Wood, Cleated-Plywood. |
| PPP-B-621 | Boxes, Wood, Nailed and Lock-Corner. |
| PPP-B-636 | Box, Fiberboard. |

Military

| | |
|---|---|
| MIL-L-10547 | Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible. |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-129 | Marking for Shipment and Storage. |

FSC 6630

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(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document froms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM Designation C148-59T Tentative Methods of Polariscopic Examination of Glass Containers.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia 3, Pa.)

3. REQUIREMENTS

3.1 Material. Material shall be suitable for the purpose intended, and as hereinafter specified.

3.1.1 Glass. The tubes shall be of clear, colorless, soda-lime or borosilicate glass, free of cracks, chips, seeds, and scratches. Any ring strain shall be not greater than temper No. 1, when tested as specified in 4.4.1. Longitudinal strain, if present, shall be faint and highly diffused.

3.1.2 Metal parts. All metal parts, except steel balls, shall be of polished brass, lacquered to protect against corrosion. The balls shall be of finely polished corrosion-resisting steel.

3.1.3 Wood case. The carrying case shall be made of a suitable hardwood, polished and varnished.

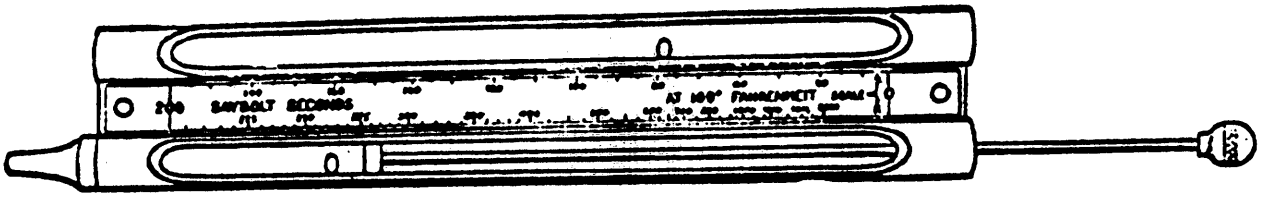
3.2 Style and design. The viscosity comparator shall be a pocket type, approximately 9 inches long, 1-1/2 inches wide, and 1/2 inch in depth, and shall be essentially as shown on figure 1. The viscosity comparator shall consist of a master reference tube containing an inert oil of known viscosity, and a steel ball. The sample tube shall contain a steel ball, and a built-in syringe, for drawing a sample of the oil to be tested. Viscosity shall be read directly from a scale in Saybolt seconds at 100° F.

3.2.1 Reference tube. The reference tube shall be of glass, encased in a brass sheath, the sheath being cut away to form a window along the tube, as shown in figure 1. The tube shall contain an inert oil of 95 viscosity index, and a spherical ball free to travel the length of the tube.

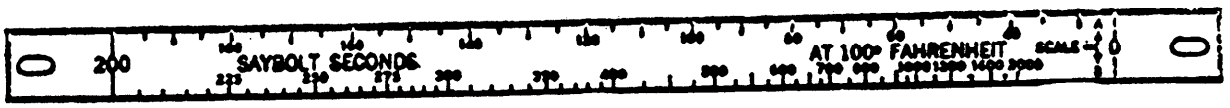


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MODEL "2" REFERENCE TUBE APPROXIMATELY TWO-THIRDS SIZE




SAMPLE TUBE




VISCOSITY COMPARATOR

FIGURE 1



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3.2.2 Sample tube. The sample tube and its sheath shall have the same dimensions as the reference tube. One end of the tube shall have a nozzle, and the other end shall be fitted with a gland or other means to form a snug-running fit with the plunger rod without any leakage. The tube shall have a ball similar to the ball in the reference tube.

3.2.3 Plunger rod and plunger. The plunger rod shall be rigid, smoothly polished, fitted on the external end with a ball or other means to serve as a handle. The internal end of the rod shall have a piston or plunger of a material that shall freely and snugly travel in the tube and be chemically inert to the oils under test.

3.2.4 Steel balls. The steel balls contained in the reference and sample tubes shall be of the same size and material, and shall have a smooth surface.

3.2.5 Scale. The graduated scale shall run parallel with, and adjacent to, the reference and sample tubes. Numerals shall be stamped or etched in clear brass on a black background on the scale. The lettering and graduations shall be distinct, and ranged for close reading from 200 to 2000 Saybolt seconds. The major graduations shall be longer than the minor graudations and shall not wash out when tested as specified in 4.4.2. The scale shall be calibrated in Saybolt seconds at 100° F., from 25 to 200 in increments of 5, from 200 to 300 in increments of 5, from 300 to 400 in increments of 10, from 400 to 700 in increments of 25, from 700 to 800 in increments of 50, from 800 to 1000 in increments of 25, from 1000 to 1200 in increments of 100, from 1200 to 1600 in increments of 150, and from 1600 to 2000 in increments of 200.

3.2.6 Carrying case. The tubes shall be nested in a draft-free carrying case. The case shall measure approximately 11 inches by 2 inches by 1-1/2 inches deep. The case shall have brass hinges and a brass holding clasp for the cover. The cover shall have a glass panel, through which the scale can be read.

3.3 Identification marking. The viscosity comparator shall be permanently and legibly marked with the name or registered trademark of the manufacturer.

3.4 Instructions. Each viscosity comparator shall be furnished with instructions and charts for use and maintenance.

3.5 Workmanship. Workmanship shall be first class throughout. Viscosity comparators shall be free from defects which detract from their appearance or may impair their serviceability.



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirments.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105.

| | Inspection level | AQL (percent defective) |
|---|---|---|
| For visual examination | | |
| Major defects | II | 2.5 |
| Minor defects | II | 4.0 |
| For dimensional examination | S-2 | 2.5 |

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105.

| Component (unit of product) | Characteristic | Requirement | Test procedure | Inspection level | AQL |
|---|---|---|---|---|---|
| Glass | Strain | 3.1.1 | 4.4.1 | S-2 | * |
| Scale | Permanency of marking | 3.2.5 | 4.4.2 | S-2 | 2.5 |

* Acceptance number shall be zero.

4.3 Examination. The viscosity comparators shall be examined to determine compliance with all requirements contained in this specification.



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4.3.1 Classification of defects. Examination shal be conducted in accordance with the following classification of defects:

Categories and defects*

Major

101 Component missing.
102 Instructions missing, or incorrect.
103 Balls not free in tubes.
104 Graduations on scale not clear.

Minor

201 Finish on components not smooth or free of rough edges.
202 Tubes not free of dirt and other foreign matter.
203 Glass not free of seeds, thin areas, and checks.
204 Plunger rod not free in tube, or leaking.
205 Identification marking on comparator missing, illegible, or incomplete.

* Examination shall not be restricted to the classified possible defects listed above.

4.3.2 Dimensional examination. The unit, including components, shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Strain. Reference and sample tubes shall be tested in accordance with ASTM Designation C148-59T.

4.4.2 Permanency of marking. The components shall be immersed in water for 1 hour, and in alcohol for 1 hour.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One (1) comparator, with case, as specified, constitutes one unit of issue.



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5.1.2 Packaging quantities. The number of units of issue indicated in table I shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table I. Packaging Quantities

| Unit package | Intermediate package | Exterior container |
|---|---|---|
| 1 unit | 12 units | 24 units |

5.1.2.1 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Each unit shall be packaged in a corrugated fiberboard folder of appropirate size and design. Closure shall be adequate to prevent spillage of contents under normal handling.

5.1.3.2 Intermediate package. Intermediate package shall be a carton of appropriate size constructed in accordance with PPP-B-636. Closure shall be adequate to prevent spillage of contents under normal handling.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type III, class 1 plywood; PPP-B-621, table III; or PPP-B-636, type CF, class weather-resistant.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.



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5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix of PPP-B-636. In addition, waterproofing shall be as specified in 30.4 of appendix to box specification.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall conform to PPP-B-601, table I, using type III, class 1 plywood; or shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 2 use; PPP-B-621, table I; or PPP-B-636, type CF, class domestic. Closure of fiberboard box shall be in accordance with method I of appendix to box specification.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES.

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Selection of applicable level of packing (see 5.2).

6.2



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6.2 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activites:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

Category: C



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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

**INSTRUCTIONS**

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION: MILITARY SPECIFICATION MIL-C-36346(DM), dated 2 December 1964
COMPARATOR, VISCOSITY, OIL, 200 - 2000 SAYBOLT SECONDS

ORGANIZATION (Of submitter) | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity) | DATE

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SUPERSEDES

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